

Unit

3

Lesson

1



Beyond Imagination.

Print the Perfect Fit.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Accurately measure their alert system components.
- Sketch an enclosure design that supports function and safety.
- Export, slice, and prepare their file for 3D printing.

Challenge questions of the lesson

- How do you ensure your component measurements are accurate before designing the enclosure?
- What tools or techniques can help minimize measurement errors?
- Why is precise measurement important for the enclosure's fit and functionality?

SEL Relation

- **SEL Relation:** Promoting Self-Management and Responsible Decision-Making: Measuring components accurately requires patience, focus, and precision, helping students develop responsibility and avoid errors that could impact the final build.
- **Enhancing Social Awareness and Relationship Skills:** Sketching an enclosure with function and safety in mind encourages students to think about the needs of the end user, share design ideas, and give constructive feedback to peers, fostering empathy and respectful collaboration.
- **Fostering a Growth Mindset and Persistence:** Exporting, slicing, and preparing files for 3D printing teaches students to embrace trial-and-error, adapt to challenges, and view mistakes as opportunities for learning and improvement.

Engage



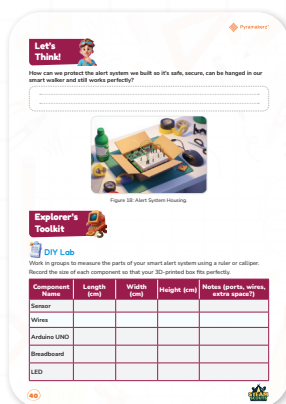
Teacher's Role:

- Guide inquiry through thoughtful questions
- Connect prior knowledge (alert system) with today's design challenge
- Spark curiosity using real-world examples
- Model empathy and active listening
- Create an inclusive, safe space for all students to share ideas

Expected from Students:

- Reflect on previous work (sensor system).
- Make observations about real-world cases/enclosures
- Respond to guiding questions thoughtfully
- Share personal ideas without fear of "wrong" answers
- Begin thinking like a designer and problem-solver

Parts of the book included.



Let's Think!



Objective:

Students will brainstorm and propose ways to safely enclose and attach their alert system onto the Smart Walker, ensuring the components remain protected, functional, and secure during real use.

Possible Answers:

How can we protect the alert system we built so it's safe, secure, and still works perfectly?

You've already created a smart alert system using a sensor and a vibration motor. But right

now, the parts are exposed—what could happen if they get bumped, dropped, or pulled by wires?

Look around you: how are electronics protected in real life? Think about your phone, your earbuds, or even your gaming controller. Why do they all have hard outer shells or cases?

- “We can design a box that fits the shape and size of the parts exactly.”
- “We need something strong to protect the electronics from breaking.”
- “The wires and sensors need to stick out, so we should leave holes for them.”
- “If we don’t cover the parts, they could fall out or stop working.”
- “A 3D printed box would let us make something that’s the perfect fit.”
- “The box has to protect, but also let us open it if we need to fix something.”
- “We can use a tool like Tinkercad to build a custom design for our system.”=

Possible Strategies for Implementation:

1. Facilitated Group Brainstorm

- Invite students to suggest ways to protect and mount the alert system.
- Use prompts like “How can we make it safe from damage?” or “What would help it stay in place while moving?”

2. Think-Pair-Share

- Give students a minute to think about enclosure ideas individually, then discuss with a partner, and finally share with the whole class.
- This helps generate a wider variety of solutions and supports participation from all learners.

3. Use Visual and Physical Examples

- Show photos or videos of safe, well-mounted electronics enclosures.
- Pass around sample materials (cardboard, plastic, 3D-printed parts) for students to handle and discuss suitability.

4. Encourage Functionality and Safety Awareness

- Ask students to imagine what could go wrong if the enclosure is not secure or safe.
- Guide them to consider user safety, component protection, and ease of use.

5. Provide Design Prompts and Templates

- Offer sentence starters like “My design will keep the alert system safe by...”
- Use simple sketch templates or 3D modeling outlines to help them visualize ideas.

6. Differentiate for Diverse Learners

- Allow responses through sketches, oral explanations, or physical mock-ups.
- Pair students with complementary skills—e.g., one strong in measuring, another in creative sketching.

Explorer's Toolkit



Teacher's Role:

- Facilitate, not lecture—ask guiding questions instead of giving answers
- Model careful measuring techniques and double-checking
- Create an inclusive, supportive atmosphere where students feel safe to ask questions and take their time
- Offer flexible tools and supports based on learning preferences and needs

Expected From Students:

- Work respectfully and collaboratively in small groups
- Measure each component (sensor, vibration motor, wires, board, etc.) using a ruler or caliper
- Record accurate dimensions: length, width, height, and any notes (e.g., "needs cutout for wires")
- Double-check their measurements for accuracy
- Ask questions if unsure—mistakes are part of learning

Parts of the book included

Figure 1b: Alert System Housing.



 **DIY Lab**
 Work in groups to measure the parts of your smart alert system using a ruler or calliper. Record the size of each component so that your 3D-printed box fits perfectly.

Component Name	Length (cm)	Width (cm)	Height (cm)	Notes (ports, wires, extra space?)
Sensor				
Wires				
Arduino UNO				
Breadboard				
LED				

Reflect

Which part was the hardest to measure? Why?

How would a small mistake in measurement affect your design?

Is there any electronic component that needs to be out of the box enclosure? what will you do?



DIY Lab

Objective:

Students will accurately measure and record the dimensions of smart alert system components, analyze the spatial needs of each part, and reflect on how measurement accuracy impacts the design of a protective enclosure for their Smart Walker system.

Answers (Sample)

(Measurements will vary depending on actual components; these are example values)

Component Name	Length (cm)	Width (cm)	Height (cm)	Notes (ports, wires, extra space?)
Sensor	4.0	2.0	1.5	Needs front opening for detection
Wires	10.0	0.2	0.2	Requires extra space for bending
Arduino UNO	7.0	5.5	1.5	Ports on sides for USB & power
Breadboard	8.0	5.5	1.0	Space for jumper wires
LED	0.5	0.5	1.5	Tip must be visible outside box

Reflect (Sample Responses):

- Hardest part to measure: Wires — they are flexible and don't have a fixed shape.
- Effect of small mistakes: Parts may not fit, causing enclosure to be too tight or too loose.
- Component outside enclosure: LED or sensor — drill/leave an opening in the box so they can work properly.

Materials Needed:

- Ruler or caliper
- Smart alert system components (sensor, Arduino UNO, breadboard, LED, wires)
- Paper & pencil for recording measurements

Activity Instructions:

1. Group Setup – Divide students into small groups, each with a complete alert system setup.
2. Identify Components – Review each part so students know what they are measuring.
3. Measure Carefully – Use a ruler or caliper to measure length, width, and height of each component.
4. Add Notes – Include extra space needed for ports, wires, or ventilation.
5. Record Results – Fill in the table accurately.
6. Reflect – Answer the reflection questions to connect measurements to real-world design considerations.

Possible Strategies for Implementation:

1. Facilitated Group Demonstration

- Teacher models how to measure one component with a caliper, explaining accuracy importance.

2. Think-Pair-Share

- Students first guess measurements, then measure, compare, and discuss differences.

3. Use Visual Guides

- Show diagram of each part with labeled measurement points.

4. Encourage Real-World Thinking

- Ask: “What happens if your 3D-printed box is 1 mm too small?”

5. Provide Measurement Checklists

- Checklist reminds students to measure all three dimensions and account for extra space.

6. Differentiate for Diverse Learners

- Allow some students to focus on sketching components while others measure.

Explain



Teacher's Role:

- **Set Purpose:** Briefly explain that slicing is the process that prepares a 3D design for the printer.
- “Before we print your box, we need to ‘slice’ it—this turns your design into instructions the printer can follow.”
- **Facilitate Discussion:** Pause at key moments to ask,
 - “What do you think infill means?” or “Why is the layer height important?”
- **Demonstrate (if possible):** Show the Cura interface live while playing the video or replicate the steps in front of the class.
- **Support Tech Navigation:** Guide students if they are using Cura or another slicer for the first time.

Expected From Students:

- Watch the video carefully to understand what slicing does and how it affects print time and quality.
- Take note of key slicing settings:
 1. Infill
 2. Layer height
 3. Supports
- Save to SD card
- Discuss or answer:
 - “Why do we need slicing before printing?”

- “What would happen if we skipped this step?”
- practice slicing a simple file or observe the teacher slicing the smart box STL file for printing.

Parts of the book included.



Screen Lessons

Click on the link to learn how you slice your design for print.

<https://youtu.be/cUVEoGfN45g?si=q-x6D9OMDJNKE38z>



Screen Lessons

Objective:

Students will understand the purpose of slicing a 3D model before printing, recognize the key slicing settings (like infill, layer height, and supports), and prepare their smart alert box design for export using slicing software to ensure a successful print.

Video link: <https://youtu.be/cUVEoGfN45g?si=q-x6D9OMDJNKE38z>



Possible Strategies for implementation:

1. Pre-Watching: Set a Purpose

Strategy: “Why Slice?”

Ask:

- “Why do you think we can’t just send the 3D model straight to the printer?”
- “What might go wrong if we don’t prepare the file first?”

Briefly explain: “Slicing turns our design into a special file with instructions for the printer—kind of like a recipe it can follow.”

2. During Watching: Active Engagement

Strategy: “Spot the Settings”

Provide students with a short checklist:

- What does infill mean?
- What is layer height?
- Why might we need supports?
- How do we save the final file?

Encourage them to take notes or raise hands when they spot these settings in the video.

3. Post-Watching: Discussion and Reflection

Strategy 1: “Setting Up for Success”

Ask:

- “What slicing setting affects how strong your box is?”
- “What would happen if we forgot to add supports?”

- “How does slicing help us avoid mistakes before printing?”

Strategy 2: “Simulate a Slice”

Open Cura or your slicing software and demonstrate slicing a sample STL. Let students suggest settings (e.g., “Let’s try 20% infill—what changes?”).

Strategy 3: “Real-World Analogy”

Compare slicing to chopping a 3D object into layers like slicing bread. Ask:

- “Why do we need to slice it into layers?”
- “What happens at the printer when it gets these sliced layers?”

Elaborate



Teacher’s Role:

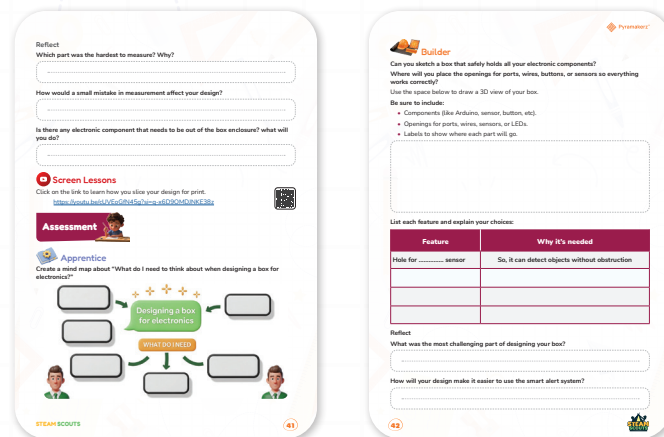
- Guide student thinking through open-ended questioning
- Encourage multiple ways of expressing understanding (drawing, writing, verbal sharing)
- Promote inclusive participation and collaborative idea generation
- Support emotional safety by affirming all contributions as valid and valuable
- Facilitate connection between prior learning (measurement, Tinkercad, 3D printing) and new planning skills (design features)
- Guide the planning process by helping students visualize how design connects to function.
- Prompt critical thinking through open-ended questions like:
 - “Where should the wires come out?”
 - “What happens if the button is hard to reach?”
- Model or demonstrate a simple example sketch, emphasizing layout and labels.
- Support all learners by offering sketch templates, physical component samples, or simplified shapes for students who need extra help.
- Encourage a growth mindset, reminding students it’s okay to revise and improve their sketch.
- Facilitate sharing through pair discussions, peer feedback, or a quick gallery walk.

Expected From Students:

- Brainstorm ideas about what goes into designing a box for electronics
- Organize their ideas into a mind map (individually, in pairs, or groups)
- Consider key areas: size, shape, access points, protection, material, printing process
- Share and reflect on their designs and ideas with peers
- Use this map later as a reference for their actual box design in Tinkercad

- Create a sketch of their enclosure that includes internal components and external openings.
- Label the sketch to show the purpose of each feature (e.g., “USB port here” or “hole for LED light”).
- Think critically about part placement, spacing, and usability.
- Use their measurements to inform size and layout.
- Reflect on how their sketch will help them build the design in Tinkercad.

Parts of the book included.



Apprentice

Objective:

By the end of this activity, students will be able to identify key factors in designing a box for electronics, such as size, access, and materials. They will create a mind map to organize their ideas and consider real-world needs in their design. Students will also reflect on how their design decisions impact the 3D printing process.

Possible Answer:

1. Component Size & Shape

- Measurements (L/W/H)
- Space between parts
- Orientation

2. Access & Openings

- Ports (USB, power)
- Sensor openings
- Buttons or switches
- LED visibility

3. Material:

- Plastic (PLA)

4. Assembly

- Hinges or removable lid
- Screws vs. snap fit

5. 3D Printing Requirements

- STL file
- Slicing settings
- Add supports if needed

Activity Instructions:

1. Introduce the Task – Tell students they will create a mind map to organize important considerations when designing a box for electronics.
2. Brainstorm as a Class – Ask: “What do we need to think about before making the box?” Record initial ideas on the board.
3. Fill the Mind Map – In groups or individually, students write one idea in each empty box. Ideas can include:
 - Size and measurements
 - Safety and protection
 - Accessibility to ports and wires
 - Material choice
 - Ventilation and cooling
4. Review & Share – Students present their completed mind maps to the class.
5. Reflect – Encourage students to think about which considerations are most important for their own project.

Possible Strategies for Implementation:

1. Facilitated Class Brainstorm

- Write all student ideas on the board before they start filling in their own mind map.

2. Think-Pair-Share

- Let students first write down their own ideas, discuss with a partner, and then share with the group to expand their thinking.

3. Use Visual Examples

- Show pictures of well-designed and poorly designed electronic enclosures to spark ideas.

4. Prompt Critical Thinking

- Ask guiding questions like:
 - “How will you make sure your box fits all the parts?”
 - “What happens if your box doesn’t have ventilation?”

5. Encourage Creativity

- Allow students to use colored pens, sketches, or diagrams to make their mind map visually engaging.

6. Differentiate for Diverse Learners

- Provide a word bank for students who may struggle with vocabulary.
- Offer sentence starters like “My box needs to...” or “I must make sure...”



Builder

Objective:

- Design an enclosure that safely holds all electronic components while allowing full functionality.
- Plan and visualize the placement of components, ports, wires, buttons, and sensors in a 3D sketch.
- Justify design choices by explaining the purpose of each feature in the enclosure.

Answers (Sample)

Features Table (Examples)

Feature	Why it's needed
Hole for ultrasonic sensor	So it can detect objects without obstruction
Slot for USB port	To allow programming and power connection without opening the box
Ventilation holes	To prevent overheating of components
Removable top cover	For easy access during repairs or upgrades
LED opening	So light signals are visible outside the box

Reflection (Sample Responses)

- Most challenging part: Fitting all components into the box while keeping it compact and easy to mount.
- How the design helps: Clear openings for sensors, LEDs, and ports make the smart alert system easy to use and maintain without dismantling the enclosure.

Activity Instructions:

1. Review Requirements – Remind students that their design must protect components, allow all functions to work, and be easy to mount on the walker.
2. Sketch the Box – In the provided space, draw a 3D view including all components, openings, and labels.
3. List Features – Fill in the table with each design feature and explain why it's necessary.

4. Reflect – Answer the reflection questions about challenges and usability.
5. Share & Discuss – Groups present their designs and receive peer feedback on functionality and safety.

Possible Strategies for implementation.

1. Model an Example Design

- Teacher sketches a sample box on the board, showing how to label parts and openings.

2. Think-Pair-Share

- Students first sketch individually, then discuss with a partner to refine ideas before finalizing.

3. Use Physical References

- Show real electronics enclosures or 3D prints to inspire placement and design ideas.

4. Prompt Critical Thinking

- Ask guiding questions like:
 - “How will you ensure your components don’t move inside the box?”
 - “What could happen if you forget ventilation holes?”

5. Encourage Creativity

- Allow use of color-coding in sketches (e.g., red for power, green for sensors).

6. Differentiate for Diverse Learners

- Provide pre-drawn box templates for students who struggle with 3D drawing.
- Allow use of Tinkercad or similar software for students more comfortable with digital design.

Elaborate



Teacher's Role:

- **Facilitate & Guide:**
 - Remind students of measurement data collected earlier.
 - Review requirements (fit, safety, function, correct openings).
 - Demonstrate exporting a Tinkercad file and basic slicing in Cura.
- **Monitor Progress:**
 - Walk around and provide feedback on design choices, ensuring openings align with component positions.

- **Prompt Reflection:**

- Encourage deeper thinking about design choices, possible improvements, and slicing considerations.

Expected from Students:

- Produce a complete 3D box design in Tinkercad meeting all constraints.
- Successfully export and slice the design in Cura, ready for 3D printing.
- Answer all reflection questions thoughtfully, showing understanding of both design and printing process.

Parts of the book included.

Showcase
Design your own custom 3D box in Tinkercad to hold the alert system
Constraints: Make sure that it holds your electronics and includes the right openings for wires, sensors, LEDs, and buttons.
Next, open Cura and prepare your design for 3D printing.
Reflect
1. What did you find easiest when designing in Tinkercad?

2. What part of your box are you most proud of? Why?

3. If you had more time, what would you add or improve?

4. What slicing setting do you think is most important for your design—and why?



Objective:

- Apply 3D design skills in Tinkercad to create a custom box for the alert system.
- Ensure the design includes correct openings for sensors, LEDs, wires, and buttons.
- Export and prepare the design in Cura for 3D printing, considering appropriate slicing settings.
- Reflect on design strengths, challenges, and improvements.

Sample Answers (Reflection)

- Easiest part: Arranging the basic box shape and placing holes in Tinkercad.
- Proudest part: Designing a compact, neat box with all ports aligned perfectly.
- If more time: Add better ventilation or smoother rounded edges for comfort and safety.
- Most important slicing setting: Infill percentage — affects the box's strength and weight.

Activity Instructions:

1. Review Measurements & Sketches – Students open their earlier sketches and measurement data.
2. Design in Tinkercad – Create the 3D model with exact dimensions, adding openings and labels.
3. Export to Cura – Save the file as .STL, import into Cura, and adjust slicing settings.

4. Check & Save – Ensure the slicing preview matches intended design, then save for printing.
5. Reflect – Answer all four reflection questions to analyze design process and outcomes.

Possible Strategies for Implementation:

1. Demonstrate Step-by-Step

- Show live how to create basic shapes, group holes, and align parts in Tinkercad.

2. Think-Aloud Technique

- As you design, explain why you make certain choices (e.g., “I’m placing this hole here so the USB cable can plug in without bending.”).

3. Peer Review

- Before exporting, students swap designs with a partner to check if openings and size look correct.

4. Real-World Connection

- Show an example of a poorly designed enclosure and ask students how they would fix it.

5. Scaffolded Support

- Provide templates for students who need extra help, while allowing advanced students to create from scratch.

6. Encourage Iteration

- Allow time for students to make small improvements after initial feedback before exporting.

Final Prototype:



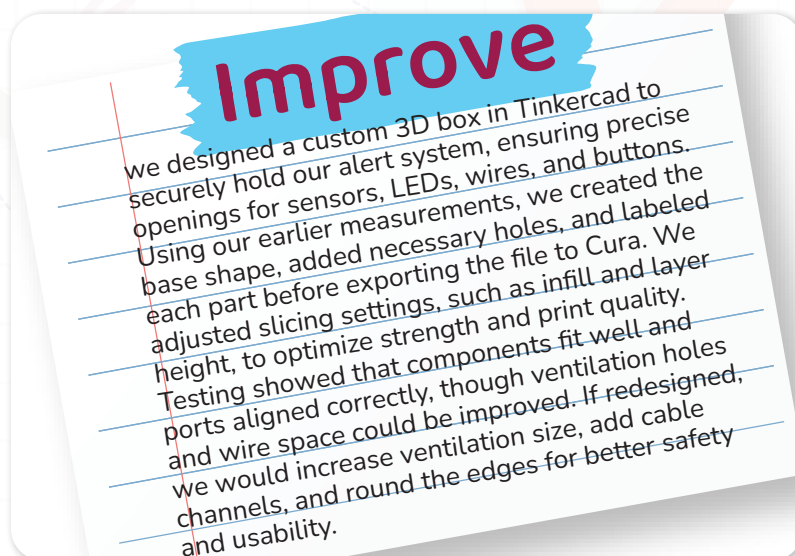
Final Prototype video:



Showcase Activity Rubric:

Criteria	Excellent (4)	Proficient (3)	Developing (2)	Beginning (1)
Design Accuracy & Functionality	All components fit perfectly; openings for sensors, LEDs, and ports are accurately placed and functional.	Most components fit; openings are mostly accurate with minor adjustments needed.	Some components do not fit; openings misaligned or incomplete.	Many components do not fit; design fails to consider placement or functionality.
Use of Tinkercad Tools	Demonstrates advanced use of shapes, grouping, alignment, and hole tools to create a polished design.	Uses Tinkercad tools effectively with few errors; design is mostly neat.	Uses only basic tools; design lacks precision or detail.	Limited use of tools; design is incomplete or poorly structured.
Slicing Preparation in Cura	Correct file export; slicing settings optimized for strength, weight, and printability; clear understanding of chosen settings.	Correct file export; slicing settings appropriate with minor adjustments needed.	File exported but slicing settings show minimal consideration for print quality.	File not exported correctly or slicing settings inappropriate for design.
Creativity & Innovation	Design shows original ideas, aesthetic consideration, and problem-solving beyond requirements.	Design includes some creative touches while meeting requirements.	Design meets basic requirements with minimal creativity.	Design is incomplete and lacks creative thought.
Reflection & Justification	Reflection answers are detailed, thoughtful, and justify design and slicing choices with clear reasoning.	Reflection answers are complete with reasonable justifications.	Reflection answers are brief or lack strong justifications.	Reflection answers are incomplete or missing.

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Now I Can

Objective:

- Encourage students to self-assess their learning and skills gained throughout the project.
- Reinforce understanding of sensor comparison, component selection, Arduino coding, and alert system integration.
- Promote metacognition by having students reflect on their progress and achievements.

Activity Instructions:

1. Review Lesson Goals – Display the “Now I Can” list to the class.
2. Self-Check – Ask students to read each statement and decide if they can confidently say “Yes” to it.
3. Evidence of Learning – For each statement, have students briefly note or discuss an example from their own work that proves they achieved it (e.g., “I tested both sensors and chose ultrasonic because it was more accurate in my trials”).
4. Peer Share (Optional) – In pairs, students share one skill they are most proud of and explain why.
5. Teacher Feedback – Provide quick verbal feedback or encouragement to confirm their progress and address any gaps.

Relation to Standards

1. ISTE Standards for Students

- **1. Empowered Learner:** Students take active roles in choosing and justifying the most suitable sensor for the Smart Walker.
- **4. Innovative Designer:** Students design, build, and refine an obstacle-detection system using Arduino and 3D-printed components.
- **5. Computational Thinker:** Students apply logical problem-solving in coding the system and integrating Bluetooth and vibration motor alerts.
- **6. Creative Communicator:** Students present their designs and explain their technical choices to peers.

2. CSTA K–12 Computer Science Standards

- **2-AP-11:** Create programs that use variables to store and modify data.
- **2-AP-12:** Modify, remix, or incorporate portions of existing programs into one's own work to develop something new.
- **3A-AP-17:** Decompose problems and subproblems into parts to facilitate design, implementation, and review.

3. Australian Curriculum – Digital Technologies

- **ACTDIP039:** Plan, manage and evaluate interactive solutions, considering safety and sustainability.
- **ACTDIP040:** Implement modular programs using an object-oriented programming language.

4. Cambridge ICT Starters

- **Level 2 – Unit 204:** Programming and Control: Create, test, and debug programs to solve real-world problems.
- **Level 2 – Unit 205:** Multimedia for Communication: Select and use appropriate tools to present technical work clearly.

5. SDG Connection

- **SDG 3 – Good Health and Well-being:** Supports the mobility and safety of individuals through assistive technology.
- **SDG 9 – Industry, Innovation, and Infrastructure:** Encourages innovation in designing practical, technology-driven solutions.